Companion Module: Excitable Phase-Synchronisation, Neuroendocrine Calibration, and Multiplex Triage Cascades

Coupling Reference to Core Framework (draft October 12 2026): Chapters 9 (Sections 9.3–9.7), 10 (Sections 10.1–10.5), 11 (Sections 11.1–11.5), and 12 (Sections 12.1–12.7)

State-Space Scope: Maps the micro-to-macro dynamical transition mechanisms connecting individual neuroendocrine calibration and temporal discounting within the biological state vector b(t) and internal institutional operators within a(t) to macroscopic demographic signalling (τ_{sig}), fast-clock institutional deformation (τ_{fin}), and post-transition basin nucleation ($N_0 \rightarrow N_i$).

1 Scope, Evidential Status, and the Three Distinct Macroscopic Phenomena

1.1 The Cross-Scale Dynamical Problem

Within the coupled *Dissipative Structure Theory* and *Niche Construction Theory* (DST–NCT) state-space x(t) = (b(t), m(t), a(t), u(t)), Chapters 9 through 12 establish that a human population b(t) is not a homogeneous labour stock, nor is an institutional coordinate system a(t) = (a_{rules}(t), a_{claims}(t), a_{models}(t)) an automated, non-agentic rule-set. Both peripheral citizens and internal institutional operators possess evolved neuroendocrine sensors that sample environmental predictability, buffer coverage (T_{cover} vs. T_{recur}), and care-boundary reliability (Σ_{care}), adjusting their temporal planning horizons (H) and behavioural allocations accordingly.

A central mathematical question arises at the interface between individual biological plasticity and macroscopic regime transitions: how does a heterogeneous population of organisms—each possessing different neuroendocrine gain, autonomic recovery rates, developmental calibration histories (τ_{dev}), social network topologies, and temporal discounting functions—transition from weakly correlated individual fluctuations ("demographic noise") into a macroscopic state in which behavioural and physiological trajectories become temporally aligned, mutually reinforcing, and hysteretically persistent (τ_{sig})? Simultaneously, how do bounded elite and managerial actors inside a(t) shift from long-horizon institutional stewardship (τ_{inst}) to synchronised short-term extraction and peripheral triage (τ_{fin}) without requiring an overt central conspiracy?

To maintain mathematical precision and avoid category errors when applying statistical physics to human social systems, three distinct phenomena must be separated at the outset:

1. **Correlated Exposure (Shared Shock Response):** Many individuals receive the same exogenous environmental input u(t) or material compression m(t), causing their physiological or behavioural states to covary in time even in the complete absence of social interaction (K = 0).
2. **Discrete State Transition or Threshold Cascade:** Many individuals cross a binary or bistable behavioural threshold—often irreversibly or hysteretically—due to cumulative network adoption or fold bifurcations, without exhibiting an ongoing oscillatory phase relationship (Granovetter, 1978; Watts, 2002; Brock & Durlauf, 2001; Scheffer et al., 2009).
3. **True Dynamical Phase Synchronisation:** Latent or observable recurrent trajectories acquire stable phase or frequency relationships across individuals or strata, measurable via a macroscopic order parameter R(t), cross-spectral coherence, phase-locking value, or contraction of conditional phase differences (Kuramoto, 1975, 1984; Shinomoto & Kuramoto, 1986; Teramae & Tanaka, 2004).

Treating a one-time population-wide drop in fertility, investment horizon, or institutional compliance as "classical Kuramoto synchronisation" merely because it occurs concurrently would be a category error. In real socio-ecological transitions, human populations combine all three processes across different timescales. This companion module formalises the exact mathematical bridge connecting excitable phase dynamics, continuous threshold cascades, and game-theoretic basin shifts, and specifies the parsimony constraints required to interface them with the core DST–NCT framework without over-parameterisation.

1.2 The Minimal Coupled State-Space Formulation

In standard mathematical physics, the classical Kuramoto model ($d\theta_i/dt = \omega_i + (K/N)\,\Sigma_j \sin(\theta_j - \theta_i)$) describes ensembles of self-sustained, continuous periodic oscillators with distributed natural frequencies ω_i. Human neuroendocrine stress recalibration, commitment monitoring, and fiduciary defection, however, are episodic, excitable, or threshold-governed rather than smoothly sinusoidal.

The established mathematical bridge between resting/thresholded systems and Kuramoto phase-locking is the **Shinomoto–Kuramoto Active Rotator** (Shinomoto & Kuramoto, 1986; Sakaguchi & Kuramoto, 1986), equivalent via standard coordinate transformation to the canonical **Ermentrout–Kopell Theta-Neuron / Saddle-Node on Invariant Circle (SNIC)** reduction of Type-I excitability (Ermentrout & Kopell, 1986; Kawamura, Nakao & Kuramoto, 2011). When augmented with **second-order dynamical inertia** (Tanaka, Lichtenberg & Oishi, 1997; Olmi et al., 2014) and coupled to slow biological calibration and horizon variables, the minimal stochastic system for agent i takes the form:

$$
\begin{aligned}
&\text{(1) Excitable Inertial Phase / Monitoring Coordinate:}\\
&m_i\, d^2\theta_i/dt^2 + \gamma_i\, d\theta_i/dt = \omega_i - a_i(S_i)\sin(\theta_i)\\
&\qquad\qquad + (K_i / k_i)\,\Sigma_j\, A_{ij}\sin(\theta_j - \theta_i - \alpha_{ij})\\
&\qquad\qquad + \beta_i\, E(t) + \sigma_{c,i}\,\xi_c(t) + \sigma_i\,\xi_i(t)\\
&\\
&\text{(2) Slow Neuroendocrine \& Allostatic Calibration } (\tau_{dev},\ \tau_{sig}):\\
&\tau_{c,i}\, dc_i/dt = F_c(E(t),\ \theta_i,\ L_i,\ \text{developmental history}) - c_i\\
&\\
&\text{(3) Effective Temporal Horizon \& Behavioural Allocation Parameter:}\\
&\tau_{q,i}\, dq_i/dt = F_q(\theta_i,\ c_i,\ q_i,\ R_X(t))
\end{aligned}
$$

Where the state coordinates and parameters map directly to biophysical and institutional processes:

- $\theta_i \in [0, 2\pi)$: The phase coordinate of an agent's recurrent **action–monitoring–commitment cycle** or episodic autonomic/HPA stress–recovery excursion (not a literal mechanical pendulum).
- ω_i: Intrinsic baseline drift or monitoring rate across the cycle.
- $a_i(S_i) = a_{0,i} + \chi_i\, S_i(t)$: The **safety-dependent pinning/restoring parameter**, representing the depth of the homeostatic potential well that holds the agent at rest when credible environmental and relational safety cues $S_i(t)$ are present.
- m_i and γ_i: Second-order **dynamical inertia** (ontogenetic calibration inertia $\tau_{dev,}$ cognitive normality schemas, and institutional commitment lock-in) and damping coefficient (in the overdamped limit $m_i \to 0$, $\gamma_i = 1$, recovering the first-order active rotator).
- $A_{ij,}$ $k_{i,}$ $K_{i,}$ α_{ij}: Adjacency matrix, node degree, social coupling gain, and phase lag across multiplex interpersonal, kin, mating, and organisational networks.

- $E(t)$ and β_i: Shared common-mode environmental forcing generated by structural cannibalisation ($dB/dt < 0$, $D_s > G_s + R_s$, Σ_{care} erosion) and the individual's **environmental sensitivity / susceptibility gain** (β_i).
- $\xi_c(t)$ and $\xi_i(t)$: Unit-variance common stochastic noise ($\sigma_{c,i}$, shared macro-volatility) and private idiosyncratic noise (σ_i, local micro-fluctuations with $Cov(\xi_i, \xi_j) \approx 0$).
- $c_i(t)$: Slowly adapting neuroendocrine and allostatic calibration profile (HPA/SAM reactivity, autonomic baseline).
- $q_i(t)$: Downstream behavioural allocation parameter governing effective temporal discount rate (1/H), fiduciary restraint, long-horizon credential investment, or reproductive timing.

Evidential Boundary: Equations (1)–(3) constitute a novel cross-scale synthesis. Its mathematical building blocks (active rotators, SNIC bifurcations, inertial hysteresis, common-noise Lyapunov contraction, and multiplex chimera states) and its biological building blocks (Differential Susceptibility, the Adaptive Calibration Model, Generalized Unsafety Theory of Stress, and dyadic physiological synchrony) are established in peer-reviewed literature. Their integration into a unified socio-ecological transition model is a proposed mechanistic mapping subject to the falsifiable empirical signatures specified in Section 6.

2 The Mathematical Bridge: From Pinned Excitability to Macroscopic Phase-Locking

2.1 Safety Pinning, SNIC Bifurcations, and Phase Excursions

In the overdamped active-rotator reduction ($\gamma_i \, d\theta_i/dt = \Omega_i - a_i \sin(\theta_i)$, where $\Omega_i = \omega_i + \beta_i E(t) + \text{social field}_i$ represents total effective drive), the qualitative behaviour of the system depends on the dimensionless ratio $|\Omega_i / a_i|$ (Shinomoto & Kuramoto, 1986):

- **Pinned Homeostatic Regime ($|\Omega_i| < a_i$):** The unit possesses two equilibria on the circle: a locally stable resting fixed point at $\theta_s = \arcsin(\Omega_i / a_i)$ and an unstable saddle/excitation threshold at $\theta_u = \pi - \theta_s$. So long as the safety-pinning potential a_i exceeds effective drive Ω_i, the organism does not oscillate continuously; it remains pinned near θ_s (homeostatic coupling), returning to baseline after minor perturbations.
- **The Distance-to-Threshold Margin (δ_i):** The effective barrier separating an individual from an unbuffered stress or horizon excursion is $\delta_i(t) = a_i(S_i) - [\omega_i + \beta_i E(t) + \text{social field}_i(t)]$.
- **Saddle-Node on Invariant Circle (SNIC) Bifurcation ($\Omega_i \to a_i^-$):** When rising common-mode environmental compression $\beta_i E(t)$, eroding safety cues $a_i(S_i) \downarrow$, or social coupling pushes $\delta_i(t) \to 0$, the stable resting state θ_s and unstable threshold θ_u collide and annihilate in a saddle-node bifurcation (Ermentrout & Kopell, 1986; Kawamura et al., 2011). In the normal form of the theta-neuron ($d\theta/dt = 1 - \cos(\theta) + (1 + \cos(\theta)) I(t)$), crossing $I = 0$ transitions the element from an excitable resting state into large phase excursions or sustained fast-clock cycling ($I > 0$). Near the SNIC boundary, dwell times lengthen and phase-response sensitivity becomes highly nonlinear, making small shared perturbations disproportionately powerful in aligning trajectories across individuals.

2.2 Common-Noise Synchronisation vs. Endogenous Peer Coupling

A critical empirical question in Chapter 9.5 is how distributed individuals align their stress and commitment trajectories *before* overt peer-to-peer imitation or organised collective action occurs. Dynamical systems theory identifies two distinct, sequentially interacting routes to macroscopic coherence $R(t) = |N^{-1} \Sigma_j \exp(i\theta_j(t))|$:

1. **Common-Mode Stochastic Synchronisation (The Teramae–Tanaka Mechanism):** Even when direct social coupling is negligible ($K \approx 0$), shared environmental fluctuations ($\sigma_{c,i}\ \xi_c(t)$ or $\beta_i\ E(t)$) can phase-lock uncoupled heterogeneous units. Teramae & Tanaka (2004) proved that weak additive common noise driving an ensemble of limit-cycle or excitable oscillators with phase-response curve $Z(\theta)$ yields a transverse Lyapunov exponent $\lambda_\perp = -(\sigma_c^2 / 2) \int [Z'(\theta)]^2\ P_0(\theta)\ d\theta < 0$. Because $\lambda_\perp$ is strictly negative, conditional phase differences between trajectories contract exponentially toward a common rhythm determined by the shared environmental forcing (Goldobin & Pikovsky, 2005; Nakao, Arai & Kawamura, 2007). In human niches, shared structural cannibalisation—simultaneous housing inflation, real-wage compression, transit/healthcare queue jitter, and τ_{fin} contract precarity—acts as this common forcing field, contracting the phase distribution of the susceptible sub-population prior to direct social contagion.
2. **Endogenous Mean-Field Entrainment (The Kuramoto–Sakaguchi Route):** Once common-mode forcing narrows the phase distribution of a subset of actors, their aggregate contribution to the social mean field $Z(t) = R(t) \exp(i\Psi(t))$ becomes non-zero. In the classical mean-field reduction ($d\theta_i/dt = \omega_i + K\ R \sin(\Psi - \theta_i)$), an individual unit locks to the collective frequency Ω_c whenever $|\omega_i - \Omega_c| \leq K\ R$ (Kuramoto, 1975; Dörfler & Bullo, 2014). Thus, common environmental forcing initiates phase contraction (R rises above noise), which then activates endogenous social coupling $K\ R$ to pull higher-threshold neighbours into the locked cluster.

2.3 Second-Order Inertia, Explosive Synchronisation, and Asymmetric Hysteresis

In a purely first-order Kuramoto model ($m_i = 0$), the transition from incoherence ($R = 0$) to coherence ($R > 0$) as coupling K increases is continuous (second-order) and fully reversible: reducing K below the critical threshold K_c immediately restores the original state. Human developmental calibration (τ_{dev}), cultural normality schemas (Chapter 9.1), and institutional rules (τ_{inst}), however, possess substantial momentum.

Incorporating the second-order inertial term $m_i\ d^2\theta_i/dt^2$ (Tanaka, Lichtenberg & Oishi, 1997; Olmi, Navas, Boccaletti & Torcini, 2014) or correlating coupling strength with node degree/frequency (Gómez-Gardeñes et al., 2011; Boccaletti et al., 2016) fundamentally alters the phase diagram:

- **First-Order Discontinuous Jump ("Explosive Synchronisation"):** High dynamical inertia m_i allows a population to resist transient environmental noise and remain in its baseline state up to an elevated forward threshold $K_{forward}$ (or forcing threshold $E_{forward}$). Once $K_{forward}$ is breached, the system undergoes an abrupt, discontinuous jump to high synchronization ($R \rightarrow 1$).
- **Asymmetric Hysteresis Loop ($K_{backward} < K_{forward}$):** Once a high-inertia cohort has been accelerated into the synchronised fast-clock basin, its angular momentum sustains coherence even if environmental stress or coupling drops well below $K_{forward}$. Restoring the baseline state requires reducing forcing below a much lower backward threshold $K_{backward} < K_{forward}$ across an extended recovery window (τ_{dev}). This supplies the formal mathematical foundation for **Demographic Hysteresis** (Chapter 10.2) and **Reverse Demographic Hysteresis** (Chapter 12.5).

2.4 Conformist vs. Contrarian Coupling and Dual-Wave Phase Splitting

A common misconception is that phase synchronisation requires the entire population b(t) to converge onto a single uniform behaviour. Hong & Strogatz (2011) demonstrated that Kuramoto networks containing a mixture of positive-coupling ("conformist" / attractive, $K_i > 0$) and negative-coupling ("contrarian" / repulsive, $K_i < 0$) oscillators, or non-zero phase lags α_{ij} (Sakaguchi &

Kuramoto, 1986; Bick, Ashwin & Rodrigues, 2016), spontaneously bifurcate into **coherent phase-separated clusters** (such as travelling waves or anti-phase π-states).

Applied to socio-ecological niches, rising common-mode stress does not collapse the population into a monolithic state; it partitions b(t) into phase-locked but directionally opposed adaptations—such as simultaneous **centrifugal withdrawal / precautionary enclaving** ($N_{i,exit}$) versus **centripetal / agonistic patch foraging** ($N_{i,capture}$), or **Homeostatic Coupling (<Care>)** versus **Allostatic Reconfiguration (<Break>)**. A naive scalar population average cancels these opposing vectors out, whereas cluster-specific order parameters (R_A, R_B, R_C) capture their simultaneous intensification.

3 Bottom-Up Biological Synchronisation: The DSM–ACM–GUTS Sensory Coupling

3.1 Parameter Mapping: Integrating DSM, ACM, GUTS, and Signal Detection Without Redundancy

Rather than stacking separate differential equations for each evolutionary and neurobiological theory, the active-rotator architecture maps four established empirical literatures directly onto distinct parameters of Equations (1)–(3):

Biological & Evolutionary Framework	**Mathematical Parameter in Equations (1)–(3)**	**Mechanistic Function and Scope Boundary**
Generalized Unsafety Theory of Stress (GUTS) *(Brosschot, Verkuil & Thayer, 2016, 2018)*	Pinning potential $a_i(t) = a_{0,i} + \chi_i S_i(t)$	Stress activation is the default, tonically disinhibited state of the organism; continuous cues of environmental and relational safety (S_i: social slack, predictable co-regulation, territorial refuge, intact Σ_{care}) maintain prefrontal (vmPFC) inhibition, holding a_i deep. Eroding safety cues flatten a_i without requiring discrete violent threats.
Differential Susceptibility & Biological Sensitivity to Context (DSM) *(Boyce & Ellis, 2005; Belsky & Pluess, 2009; Pluess, 2015)*	Environmental gain β_i and threshold margin $\delta_i(t) = a_i - \Omega_i$	Environmental sensitivity is distributed continuously across b(t). High-plasticity "orchids" occupy the high-β_i, small-δ_i tail (bi-directionally responsive to both adverse compression and supportive buffering); "dandelions" occupy the low-β_i, high-inertia m_i tail.

The Smoke Detector Principle *(Nesse, 2001, 2005)*	Asymmetric cost bias on excitation barrier δ_i	Because the evolutionary fitness cost of a missed existential or reproductive-niche failure (C_m) vastly exceeds the cost of a false alarm (C_f), the biological sensor is optimally biased toward early firing ($\delta_i \downarrow$) under ambiguous uncertainty.
Adaptive Calibration Model (ACM) & Life History Compression *(Del Giudice, Ellis & Shirtcliff, 2011; Ellis et al., 2017; Del Giudice, 2020)*	Slow neuroendocrine vector $c_i(t)$ and horizon parameter $q_i(t)$	Integrates chronic GUTS disinhibition and phase excursions θ_i across τ_{dev} and τ_{sig} to route individuals into distinct stress-responsivity profiles (*sensitive*, *buffered*, *vigilant-anxious*, or *unemotional/deactivating*), modulating temporal discounting and commitment horizons q_i (Kimura et al., 2013; Lempert et al., 2012).
Dyadic Biobehavioural & Endocrine Synchrony *(Feldman et al., 2011; Schury, Nater & Häusser, 2020; de Groot et al., 2012)*	Multiplex coupling matrix K_i A_{ij}	Grounds interpersonal physiological coupling ($K_{ij} > 0$) in verified short-lag cardiac coordination, apocrine chemosignals, and shared-identity cortisol contagion, while discarding contested Polyvagal phylogenetic/anatomical claims (Grossman, 2023).

3.2 The Two-Stage GUTS–ACM Serial Coupling and Prosthetic Safety Enclaving

A critical theoretical tension arises if GUTS and ACM are combined uncritically: GUTS posits that autonomic stress readiness is *always on by default* when safety cues are absent (operating on $\tau_{autonomic}$: seconds to hours), whereas ACM demonstrates that organisms in chronically harsh environments frequently develop *blunted* HPA and autonomic reactivity—the ACM **unemotional / deactivating (Type IV)** profile (operating across τ_{dev} and τ_{sig}: months to decades).

This tension is resolved by recognising that **GUTS and ACM are coupled in series across timescales**:

1. **GUTS as the Fast Proximal Input Gate ($\tau_{autonomic}$):** The mammalian nervous system does not primarily count statistical homicide rates or macroeconomic GDP figures; it continuously audits the immediate availability of social slack, unhurried conspecific prosody, alloparental redundancy (Σ_{care}), and controllable territorial refuge. When τ_{fin} structural cannibalisation hollows out household time buffers and degrades behavioural predictability in shared public bottlenecks, GUTS prefrontal inhibition fails chronically.
2. **ACM as the Slow Developmental and Behavioural Router (τ_{dev}, τ_{sig}):** Because no organism can survive infinite, unbuffered GUTS arousal without metabolic burnout (monotonically rising allostatic load $dL_i/dt > 0$, augmented by expressive suppression costs S_i in conflict-avoidant cultures; Gross & Levenson, 1997), the ACM switchboard routes phenotypes along two primary defensive trajectories:
    - **The Vigilant-Anxious Trajectory (ACM Type III — Prosthetic Safety Enclaving):** Disproportionately expressed among high-susceptibility cohorts ("orchids") and cohorts structurally dependent on formal institutional pipelines. Retaining acute sensitivity to GUTS disinhibition, these actors survive degraded shared spaces (N_0) by expending private capital, leverage, or regulatory effort to construct **prosthetic safety enclaves (B_{sub})** across four boundary-gating dimensions—*spatial-territorial* (sealed residential "caves"), *kinetic-corridor* (armoured SUVs, ANC headphone/smartphone sensory bunkers), *virtual-temporal* (asynchronous digital cocoons and simulated SLHS sanctuaries), and *normative-institutional* (bureaucratic speech and interaction codes)—solely to force their prefrontal safety brake ($a_i(S_i)$) back on.
    - **The Unemotional / Deactivating Trajectory (ACM Type IV — Second-Order Physiological Blunting):** Disproportionately expressed among cohorts lacking material buffers to escape exposure or facing agonistic/externalising selection pressures. Here, ACM executes a second-order physiological triage by down-regulating receptor sensitivity (β_i and autonomic gain blunted), severing empathetic co-regulation so the organism can forage or compete inside high-friction environments without suffering acute autonomic collapse.

3.3 The Vanguard Signal-to-Noise Algebra: From $O(\sqrt{N_S})$ Cancellation to $O(N_S)$ Forcing

Why does the high-plasticity "orchid" sub-population ($S \subset b(t)$, with large β_i and narrow threshold margin δ_i) register as ignorable clinical or subcultural noise during benign regimes ($dB/dt > 0$), yet function as a coherent macroscopic early-warning sensor (τ_{sig}) when structural cannibalisation begins ($dB/dt < 0$)?

Vector algebra on the complex order parameter $Z_S(t) = N_S^{-1} \Sigma_{i \in S} \exp(i\theta_i(t)) = R_S(t) \exp(i\Psi_S(t))$ establishes a three-stage sequence:

```
STAGE I: Buffered Niche (dB/dt > 0, Intact Σcare)
        Private noise (σi ξi) dominates over shared forcing (βi E(t) ≈ 0).
        Orchids experience isolated excursions, but phases θi are uniform
on [0, 2π).
        Vector sum |Σ exp(iθi)| scales as O(√NS)  ==>  Order parameter
RS(t) -> 0.
        Observable: Moderate/high individual variance VS(t), zero macro-
signal RS(t).
                                          |
        [ Onset of Structural Cannibalisation & Downward Volatility
Transfer ]
                                          ▼
STAGE II: Probing & Common-Noise Contraction (Teramae–Tanaka Route)
         Shared friction E(t) and common noise ξc(t) erode safety pinning
ai(Si).
         Negative transverse Lyapunov exponent (λ_⊥ < 0) contracts phase
differences
         across S even before peer contagion (Kij) dominates.
         Observable: VS(t) peaks, then conditional phase coherence RS(t)
rises sharply.
                                          |
        [ Superlinear Mean-Field Jump: O(√NS) Noise -> O(NS) Coherent
Signal ]
                                          ▼
STAGE III: Multiplex Propagation into the High-Inertia Majority (Layers A -
> C)
          The coherent vanguard field (KCS RS) propagates across kin,
mating, and
          common-knowledge channels, driving the majority across its
hysteresis cliff.
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In Stage III, propagation from the synchronised vanguard ($R_S \to 1$) into the lower-sensitivity, higher-inertia majority ("tulips" and "dandelions") operates across a **multiplex coupling operator** $K_{ij} = K_{kin}\, A^{kin}_{ij} + K_{mate}\, A^{mate}_{ij} + K_{peer}\, A^{peer}_{ij} + K_{media}\, A^{media}_{ij}$:

- **The Intergenerational Kin Backchannel (K_{kin} — Chapter 10.1):** Older adult cohorts do not compute abstract macroeconomic exergy balances; they continuously monitor the maturation, household formation, and reproductive progression of younger cohorts within their *Recognised Succession Field*. When phase-locked vanguard decoupling produces simultaneous milestone failures across multiple siblings, adult children, and nieces/nephews, high-reliability kin observation overrides low-plasticity schema inertia.
- **Assortative Mating and Matching Bottlenecks (K_{mate}):** Long-term pair-bonding and cooperative breeding (Hrdy, 2009) form a two-sided matching market requiring compatible horizon commitments (q_i). When a synchronised vanguard cohort (or one sex undergoing asymmetric dimorphic calibration under Σ_{care} erosion, Chapter 9.2.1) truncates its commitment horizon or withdraws into prosthetic safety enclaves, it mechanically constricts the feasible matching set for less-sensitive partners without requiring behavioural imitation.
- **Common-Knowledge Milestones (K_{media} / K_{peer} — Chwe, 2001/2013):** Visible institutional failures, school/ward closures, or public demographic announcements convert dispersed

second-order worry ("I see delays in my family") into mutual common knowledge ("Everyone knows that everyone sees the trajectory is obstructed"), triggering a Granovetter/Watts threshold cascade across the high-inertia population (Chapter 10.2).

3.4 Network Topology, Institutional Individualism, and the Dual-Function Kin Filter

The multiplex active-rotator equation reveals why the macro-level synchronisation dynamics of a population b(t) depend critically on its underlying relational topology—specifically, the historical balance between **horizontal kin/community networks** (A^{kin}_{ij}, A^{peer}_{ij}) and **vertical institutional individualism** (such as 16th-century European Protestant individualism, Anglo-American market individualism, or Swedish *statsindividualism*) (Henrich, 2020; Chapter 11.2).

In the distance-to-threshold equation for an excitable agent i,

$$\delta_i(t) = [a_{0,i} + \chi_i S_{kin}(t) + \chi_{inst} S_{inst}(t)] - [\omega_i + \beta_i E(t) + K_{kin} R_{kin}(t) + K_{peer} R_{peer}(t)]$$

kin-bonding and institutional individualism manipulate the pinning potential $a_i(S_i)$ and the social coupling matrix K_{ij} in opposite ways, creating a sharp regime-dependent trade-off across the niche cycle.

3.4.1 The Dual Role of Kin-Bonding: Micro-Co-Regulation (a_i Shield) vs. Upward Backchannel (K_{kin} Amplifier)

In human developmental biology, kin-bonding and phase-locking are not separate mechanisms; **dyadic and alloparental kin-bonding is micro-scale physiological phase-locking** (Feldman et al., 2011). High-susceptibility ("orchid") children possess high environmental gain (β_i) alongside a shallow intrinsic self-regulation well ($a_{0,i}$), rendering them unable to maintain a pinned slow-strategy baseline ($\delta_i > 0$) in isolation. However, because differential susceptibility is bi-directional (χ_i is correspondingly large; Belsky et al., 2007), embedding an orchid child within a dense, predictable kin and alloparental cradle ($S_{kin} \uparrow$) allows the child's autonomic and HPA rhythms to phase-lock onto the slow, homeostatic cycle (<Care>) of caregivers.

Consequently, intact kin networks act as a **state-dependent nonlinear filter** on macroscopic phase synchronisation:

- **The Shield Effect (Dampening Macro-Forcing When Buffered):** So long as household and kin buffers remain intact (S_{kin} high), deep local safety pinning holds $\delta_i > 0$ even when macro-environmental volatility E(t) rises. High-sensitivity individuals remain phase-locked to their local family cradle rather than locking onto macro-societal panic.
- **The Backchannel Amplifier (Upward Contagion When Breached):** Once structural cannibalisation ($dB/dt < 0$) depletes household reserves or blocks young adults from forming households after leaving the natal nest, S_{kin} can no longer prevent vanguard excursions ($R_A \rightarrow 1$). At that point, the conserved biological coupling between parents, grandparents, and offspring ($K_{kin} > 0$; Schury et al., 2020) reverses function: the kin tie becomes a high-gain transmission wire through which synchronised youth distress and obstructed milestones travel *upward* ($K_{CA} R_A$), pulling high-inertia "dandelion" elders across their hysteresis threshold (Chapter 10.1–10.2).

3.4.2 The Institutional Paradox of Individualism: Hub-and-Spoke Topology

From the perspective of a vertical administrative coordinate system a(t) in N_0, dense horizontal kin and village networks (K_{kin}, $K_{peer} \gg 0$) present an ongoing coordination obstacle: modular clan clusters retain independent phase inertia, resist individualised labour mobility, and easily convert local friction into horizontal peer-contagion rebellions, feuds, or tax resistance ($R_{peer} \rightarrow 1$).

By dismantling intensive kin networks and replacing village/clan alloparenting with direct state-to-individual contracts (*statsindividualism*) or atomised market relations, the vertical institution performs a topological surgery on b(t): it severs the horizontal coupling matrix (A^{kin}_{ij}, $A^{peer}_{ij} \to 0$) and rewires every atomised individual into a **Hub-and-Spoke (Star) Topology** connected 1-to-1 to the central institutional ledger ($S_{inst}(t)$ and $E(t)$).

Structural & Dynamical Dimension	**Kin / Community-Collective Topology (e.g., Clan, *Byalag*, or Catholic Familial Schemas)**	**Atomised Individualist Topology (e.g., 16c Protestant, Liberal Market, Swedish *Statsindividualism*)**
Coupling Architecture (A_{ij})	**Modular Horizontal Clusters:** High local K_{kin} and K_{peer}; weak direct dependence of the individual on the central vertical hub N_0.	**Hub-and-Spoke (Star Network):** $K_{kin} \approx 0$, $K_{peer} \approx 0$; every individual i is horizontally decoupled and wired directly to the central state/market bus a(t).
GUTS Safety Pinning ($a_i(S_i)$)	**Decentralised (χ_i S_{kin}):** Extended family housing, alloparental redundancy, and local parish/community ritual buffer a_i against central fiscal shocks.	**Centrally Monopolised (χ_{inst} S_{inst}):** a_i depends directly on central Σ_{care} queues, personal credit (a_{claims}), and individual labour-market return.
Orchid Dynamics Under Normal Friction	Co-regulated by local kin rhythms (<Care>) or routed into dedicated communal sanctuaries (e.g., monastic orders).	High personal autonomy during niche expansion (dB/dt > 0), but unbuffered private exposure ($O(\sqrt{N_S})$ noise) when local friction rises.
Primary Failure Mode Under Contraction (dB/dt < 0)	**Vocal Kuramoto Peer Contagion (K_{ij} Route):** Local clusters phase-lock early into visible street protest, informal clan mutual-aid (B_{sub}), and overt non-compliance, buffering the demographic floor for longer.	**Silent Teramae–Tanaka Snap ($\lambda_{\perp} < 0$ Route):** Near-zero early street protest (since $K_{peer} \approx 0$), followed by an abrupt, simultaneous common-noise collapse in fertility, youth mental health, and somatic trust once the central hub degrades.

This topological distinction explains a central comparative anomaly during niche contraction (dB/dt < 0):

- **Immunity to Horizontal Peer Rebellion:** Because $K_{peer} \approx 0$, atomised individualist polities exhibit remarkable surface calm during early structural cannibalisation. Unbuffered orchids lacking kin co-regulation either internalise their excursions as individualised psychiatric morbidity or seek synthetic co-regulation inside digital/ideological swarms and prosthetic safety caves (B_{sub}) without triggering coordinated street rebellions.
- **Hyper-Vulnerability to Teramae–Tanaka Common-Noise Contraction:** However, because institutional individualism stripped away heterogeneous kin shock-absorbers ($S_{kin} \to 0$) and wired every citizen's safety brake (a_i) directly to the same central fiscal, housing, and Σ_{care} bus, the entire susceptible sub-population experiences central institutional jitter as an identical, simultaneous common-mode forcing field ($\beta_i E(t) + \sigma_c \xi_c(t)$). By Teramae & Tanaka's (2004) theorem ($\lambda_\perp < 0$), uncoupled oscillators ($K = 0$) subjected to a strong shared forcing field contract their phases simultaneously. Consequently, highly individualised societies bypass noisy horizontal resistance only to suffer a silent, synchronised demographic and psychosomatic phase-snap (τ_{sig}) the moment the central institutional hub falters.

4 Top-Down Elite and Managerial Synchronisation ("Shared Triage" Without Conspiracy)

In parallel with bottom-up centrifugal withdrawal, Chapter 12 models **centripetal decoupling**: as net exergy surplus contracts ($dB/dt < 0$) and buffer coverage falls ($T_{cover} < T_{recur}$), operators positioned inside or adjacent to the administrative and allocative nodes of a(t) shift from long-horizon stewardship (τ_{inst} — Olson's 1993 "stationary bandit" equilibrium, Ostromian commons restraint, and North–Wallis–Weingast elite bargains) to synchronised short-horizon extraction, fiduciary defection, and peripheral governance triage (τ_{fin}).

Crucially, this collective managerial pivot requires **no central conspiracy, smoke-filled room, or teleological malice**. It emerges mechanically from four interlocking dynamical and game-theoretic processes:

4.1 High-Gain Clock Entrainment (τ_{fin} Dashboards as Common Forcing)

Unlike peripheral citizens anchored to slow material and developmental clocks (τ_m, τ_{dev}), corporate executives, financial allocators, and senior public-sector managers are structurally wired into shared, high-frequency quantitative monitoring instruments: bond yields, credit spreads, quarterly earnings, liquidity ratios, polling tickers, and New Public Management (NPM) throughput dashboards.

In phase-dynamical terms ($d\theta_i/dt = \omega_i + \varepsilon \xi_{fin}(t)$), these rapid symbolic indicators act as a **high-gain common forcing field** (Pikovsky, Kurths & Rosenblum, 1997). When systemic volatility rises or real physical surplus contracts, the gain ε of the shared τ_{fin} signal overwhelms the heterogeneous long-horizon deliberative cycles (ω_i on τ_{inst}) of individual decision-makers, entraining their operational rhythms and subjective discount rates onto the hyper-fast tactical clock (τ_{fin}) in unison.

4.2 Global-Games Equilibrium Selection and Simultaneous Defection

Why do competing insider coalitions stop compromising for the long-term reproduction of N_0 and flip simultaneously into defensive asset capture (B_{sub}) and risk-offloading onto peripheral balance sheets?

- In an institutional assurance or public-goods game (Stag Hunt), long-horizon fiduciary restraint—leaving buffers inside the common pool to fund slow infrastructure repair R_s and

universal care Σ_{care}—is an equilibrium strategy for an insider *only* so long as that actor expects rival insiders to also exercise restraint.

- **The Carlsson–van Damme / Morris–Shin Global-Games Mechanism:** Carlsson & van Damme (1993) and Morris & Shin (1998) proved that when agents in a coordination game observe an underlying fundamental state variable (here, real institutional solvency and buffer headroom $B_{available}$) with even infinitesimal private noise, the multiplicity of equilibria vanishes. The game acquires a **unique, mathematically sharp switching threshold** in belief space.
- **The Defection Cascade:** So long as noisy solvency signals remain above the critical threshold, insiders maintain stationary stewardship. The moment indicators of structural cannibalisation cross the Global-Games threshold, higher-order beliefs flip instantaneously: every rational insider calculates that other insiders are observing the same deteriorating fundamentals and are about to defect. To avoid being the last fiduciary guardian holding unbacked symbolic claims ($a_{claims}(t)$), insiders simultaneously execute a defensive run—converting decaying macro-claims into private tangible assets (B_{sub}), stripping maintenance budgets ($D_s > G_s + R_s$), and shifting to shared peripheral triage.

4.3 Intra-Hierarchical Sorting: Plasticity, Selection, and Isomorphism Across Sub-Aggregates

Within the administrative and professional strata of a(t), this synchronisation acts across three functionally distinct sub-aggregates—**Symbolic-Procedural Guardians** (civil servants, legal/regulatory compliance, academia, media), **Allocative-Financial Brokers** (executives, procurement directors, financial-legal structuring), and **Technical-Material Operators** (grid/civil engineers, logistics and clinical leads)—via three reinforcing mechanisms:

1. **In-Situ Neurocognitive Plasticity (ACM & Threat-Rigidity):** Under perceived status or resource contraction, incumbent managers experience HPA/SAM stress recalibration and organisational **threat rigidity** (Staw, Sandelands & Dutton, 1981; Chapter 13.3)—restricting information processing, tightening hierarchical control, and truncating personal planning horizons ($H \rightarrow \tau_{fin}$).
2. **Organisational Selection and Purging (Hannan & Freeman, 1977, 1984):** Invoking "strategic urgency" or "fiscal emergency," institutional hierarchies demote, sideline, or purge slow-calibrated fiduciary guardians and technical operators who insist on statutory deliberation (τ_{inst}), impartial due process, or long-term physical maintenance (R_s), replacing them with fast-clock transactional brokers (Allocative-Financial Brokers).
3. **Coercive, Mimetic, and Normative Isomorphism (DiMaggio & Powell, 1983):** Faced with shrinking headroom, separate ministries, municipalities, universities, and corporations rapidly copy identical "lean triage," outsourcing, metric-gaming, and low-exergy coercive containment templates from peer organisations to preserve credit ratings and peer legitimacy.

4.4 The Dual-Layer Chimera: Institutional Decoupling and Schema Mimicry

A defining feature of late-stage contraction in Chapter 12.2 is the **Chimeric Institution**: even after internal allocative networks have phase-locked onto short-horizon τ_{fin} triage, the institution continues to broadcast the cooperative moral vocabulary, universalist promises, and procedural ceremonies of the old growth era (N_0).

In organisational sociology, this is formalised by **Meyer & Rowan's (1977) Institutional Decoupling** ("myth and ceremony") combined with **Chapter 12.2 Schema Mimicry**:

- **Low-Exergy Legitimacy Signalling:** Maintaining legacy verbal narratives and symbolic targets in $a_{models}(t)$ ("efficiency," "modernisation," "green transition," "keeping the household

in order") carries near-zero immediate metabolic cost relative to physical infrastructure repair (R_s), yet delays common-knowledge recognition of fiduciary defection among the high-inertia public.
- **The Chimeric Dual Strategy of Symbolic Guardians (Chapter 12.2.1):** Symbolic-Procedural Guardians whose embodied capital is 100% specific to N_0 combine *overt ideological orthodoxy*—policing the declarative legitimacy of N_0 and stigmatising peripheral exit—with *covert centripetal opportunity hoarding* (Tilly, 1998; Reeves, 2017), using residential sorting, elite schooling, and private networks (B_{sub}) to insulate their own offspring's developmental window (τ_{dev}) from the decaying public substrate they officially administer.

5 The Multiplex Chimera Architecture: Coupling Vanguard, Elites, and the Core

5.1 Three-Layer Multiplex Formulation

In a complete society, the bottom-up biological cascade (Section 3) and the top-down managerial triage cascade (Section 4) unfold simultaneously on a **three-layer interpinned multiplex network**:

- **Layer A (The High-Plasticity Vanguard):** Low safety-pinning threshold $a_i^{A,}$ high environmental sensitivity $\beta_i^{A,}$ low inertia m_i^A ("orchids" and young cohorts entering τ_{dev} without legacy normality anchors).
- **Layer B (The Elite / Managerial Stratum):** Densely connected allocative and administrative insiders driven by high-gain τ_{fin} tickers ($F^B(t)$) and Global-Games solvency thresholds.
- **Layer C (The High-Inertia Core):** Low sensitivity $\beta_i^{C,}$ deep cultural schema pinning $a_i^{C,}$ and high dynamical inertia m_i^C ("tulips" and "dandelions" executing routine material maintenance R_s and institutional compliance).

For oscillator/excitable unit i in layer $X \in \{A, B, C\}$, the multiplex system is governed by (Boccaletti et al., 2014, 2016; Kachhvah & Jalan, 2021; Wu et al., 2022):

$$m_i^X \, d^2\theta_i^X/dt^2 + \gamma_i^X \, d\theta_i^X/dt = \omega_i^X - a_i^X(S_X) \sin(\theta_i^X) + \Sigma_{Y \in \{A,B,C\}} (K_{XY} / N_Y) \Sigma_{j \in Y} \sin(\theta_j^Y - \theta_i^X - \alpha_{XY}) + F^X(t)$$

5.2 Chimera Coexistence and Double Explosive Transitions Across the Niche Cycle

In nonlinear physics, an **Abrams–Strogatz Chimera State** (Abrams & Strogatz, 2004) is a macro-dynamical regime in which a coupled network spontaneously splits into coexisting coherent (phase-locked, $R \to 1$) and incoherent/pinned ($R \approx 0$) sub-populations. In multilayer networks, Kachhvah & Jalan (2021) and Wu et al. (2022) demonstrate that interpinned layers exhibit **coexisting chimera states and double explosive (first-order) synchronisation transitions**.

This multiplex architecture maps the exact chronological sequence of a DST–NCT regime transition ($N_0 \to \Sigma \to N_i$):

Transition Phase	Layer A (Orchid Vanguard)	Layer B (Elite / Managerial Nodes)	Layer C (High-Inertia Dandelion Core)	Macroscopic State-Space Consequence in x(t)

Phase I: Buffered Expansion (dB/dt > 0, $T_{cover} > T_{recur}$)	**Asynchronous Probing ($R_A \approx 0$, High V_A):** Scattered allostatic <Break> probes cancel out as harmless cultural/clinical variety; keeps evolutionary escape repertoire warm.	**Stationary Stewardship (τ_{inst} Lock):** High buffer surplus rewards slow fiduciary rules; fast τ_{fin} extraction is suppressed by monitoring overhead.	**Pinned Homeostatic Carrier (R_C Locked to <Care>):** Deep safety pinning a_i^C sustains linear work, tax compliance, and transgenerational reproduction (τ_{dev}).	**Homeodynamic Consolidation of N_0:** Physical structures s(t) and reserves B(t) expand; claims $a_{claims}(t)$ remain backed by material throughput.
Phase II: Structural Cannibalisation & Multiplex Chimera (dB/dt < 0, $\tau_{fin} \ll \tau_m$)	**First Explosive Jump ($R_A \to 1$ on τ_{sig}):** Common-mode friction erodes a_i^A; vanguard phase-locks into somatic unsafety, prosthetic enclaving, and centrifugal demographic decoupling.	**Second Explosive Jump ($R_B \to 1$ on τ_{fin}):** Global-Games threshold crossed; insiders lock into tacit shared triage and B_{sub} capture while broadcasting **Schema Mimicry** downward to keep a_i^C pinned.	**Chimeric Lag (R_C Remains Pinned via Inertia m_i^C):** Absorbs creeping physical dilution; continues executing R_s maintenance under legacy normality schemas and Parakin Placebos.	**The Chimeric Institution (Ch. 12.2):** Fast-clock coherence at both peripheral (A) and central (B) extremes coexists with a still-compliant inertial core (C).
Phase III: The Hysteretic Cliff & Basin Snap (Σ_{care} & Σ Breached)	**Pre-Compiled Escape Execution:** Deploys rehearsed cognitive/coalitional scripts (Regime A Flight Simulator) to nucleate $N_{i,exit}$ arks.	**Modular Capture ($N_{i,capture}$):** Severs fiscal/energetic ties to bankrupt macro-centre; locks down viable modules of s(t) and B(t) via low-exergy coercion.	**Delayed First-Order Snap ($\delta_i^C \leq 0$):** Combined cross-layer forcing ($K_{CA} R_A + K_{CB} R_B$) and physical Σ_{care} failure breach Layer C's saddle-node barrier,	**Macro-Basin Termination ($N_0 \to N_i$):** The over-accelerated macro-shell fractures into shallower successor basins scaled to local exergy flux.

			triggering an abrupt compliance cascade.	
Phase IV: Post-Transition Reverse Hysteresis ($T_{cover} > T_{recur}$ in N_i)	**Bi-Directional Orchid Re-Pinning:** Senses restored local predictability across τ_{dev} first; re-pins into <Care> and broadcasts successful family milestones.	**New Stationary Consolidation :** Bound by local substrate limits ($G_s + R_s > D_s$), extractive coalitions either stabilise as stationary coordinators or self-terminate.	**Hysteretic Reset Above the Minimum Care Floor:** Once shared observation of vanguard succession crosses the elevated backward threshold ($K_{backward}$), Layer C consolidates the new slow basin.	**Slow-Strategy Re-Nucleation (Ch. 12.5):** Human biological continuity (b(t)) survives the death of N_0 by re-establishing <Care> above the τ_{dev} survival floor.

6 Competing Mathematical Explanations, Parsimony Audit, and Falsifiable Signatures

6.1 Comparative Model Selection: Excitable Kuramoto vs. Threshold, Ising, and Fold Models

A rigorous scientific synthesis must specify where simpler mathematical formalisms outperform phase oscillators, and identify the exact empirical residuals that require the Excitable Kuramoto / Multiplex Chimera architecture:

Mathematical Family	**Canonical Equations & Key Literature**	**Where This Formalism Is Superior / Simpler**	**What It Misses (Why Excitable Kuramoto Is Required)**
1. Heterogeneous Threshold Cascades	$x_i(t+1) = 1\{\Sigma_j A_{ij} x_j / k_i \geq \varphi_i\}$ *(Granovetter, 1978; Watts, 2002)*	Superior for irreversible, one-time binary adoption, migration, or compliance exit cascades where only the fraction of activated neighbours matters.	Discards relative timing, recurrent decision/monitoring cycles, clock-speed entrainment ($\tau_{inst} \rightarrow \tau_{fin}$), and common-noise phase contraction prior to peer adoption.

2. Discrete-Choice / Ising & Potts Spin-Glass Models	$H = -\Sigma_{ij} J_{ij} s_i s_j - \Sigma_i h_i s_i$; $m = \tanh[\beta(h + Jm)]$ *(Weidlich, 1971, 2000; Brock & Durlauf, 2001; Castellano et al., 2009)*	Superior for modelling static equilibrium multiplicity, binary/multi-state ideological polarisation, and random-field individual heterogeneity (h_i).	Models *which static state* dominates rather than *when trajectories align in time*; cannot represent frequency entrainment or oscillatory chimera states.
3. Fold Bifurcations & Critical Slowing Down	dx/dt = f(x; μ); relaxation time `τ_{relax} ≈ 1/	λ	→ ∞asλ → 0⁻` *(Scheffer et al., 2001, 2009)*

The Null Hierarchy Rule: In empirical applications (Chapter 14), the analyst must test explanations in order of increasing complexity: Correlated Exposure (Shared Shock) → Binary Threshold / Ising Cascade → Fold / Critical Slowing Down → Excitable Phase Synchronisation, retaining the phase-synchronisation layer only where recurrent temporal structure, clock-speed entrainment, or vanguard variance-to-coherence contraction is present.

6.2 The Parsimony Lock (Minimum Viable Hybrid Architecture)

To prevent over-parameterisation ("curve-fitting" post-hoc narratives with redundant variables), the combined framework enforces three strict parsimony reductions:

1. **Subsuming GUTS and DSM into Active-Rotator Parameters ($a_i(S)$ and β_i):** GUTS does not add a separate differential equation; it defines the restoring parameter $a_i(S_i)$ holding the active rotator pinned at rest. DSM defines the continuous distribution of environmental gain β_i and inertia m_i across the population.
2. **Eliminating Contested Polyvagal Neuroanatomy:** Following Grossman's (2023, 2026) neurophysiological critiques, the model drops Porges's phylogenetic dorsal/ventral vagal hierarchy while retaining empirically verified sympathetic/parasympathetic regulation, apocrine chemosignals (de Groot et al., 2012), and dyadic cardiac/cortisol co-regulation (Feldman et al., 2011; Schury et al., 2020) as the biological substrate of K_{ij}.
3. **Unifying Global Games and Phase Locking via Threshold Gating:** Rather than running conflicting continuous-phase and discrete game-theoretic equations in parallel, the Global-Games mechanism acts as a sharp **threshold gate on coupling and horizon parameters** (q_i and K_{BB}): once noisy solvency or vanguard coherence signals ($R_A(t)$) cross the unique common-knowledge threshold, managerial and majority actors execute a first-order basin switch.

6.3 Six Discriminating Empirical Signatures (Falsifiability Programme)

To prove that a macroscopic shift in time horizons (H), fertility postponement, or managerial triage is driven by coupled phase-synchronisation rather than passive, independent reactions to a common economic shock (K = 0), longitudinal panel and network data must exhibit six specific signatures:

- **Signature 1 — The Vanguard Variance-Then-Coherence Sequence (V_A ↑ followed by R_A ↑):** In high-susceptibility cohorts (Layer A, indexed by psychometric/autonomic sensitivity or young developmental age), within-group response variance $V_A(t)$ rises first while phase coherence $R_A(t)$ remains low (the asynchronous probing regime), followed by a sharp rise in phase coherence $R_A(t) \to 1$ (contraction of conditional timing differences) *prior* to a shift in the Layer C population mean.
- **Signature 2 — Systematic Cross-Layer Lead–Lag Directionality:** Cross-spectral coherence and lagged cross-correlations must demonstrate that Layer A (τ_{sig} somatic unsafety, prosthetic gating, and commitment withdrawal) and Layer B (τ_{fin} insider liquidity hoarding and maintenance deferral) phase-lock systematically *ahead* of Layer C compliance withdrawal, rather than all strata peaking simultaneously with an exogenous shock u(t).
- **Signature 3 — Positive Residual Network Coherence ($R_{peer} > 0$):** Using geographically or sectorally staggered shocks while conditioning out shared macro-exposure E(t), significant residual alignment in horizon truncation, fertility postponement, or corporate triage must propagate along measured kin (A^{kin}_{ij}), mating (A^{mate}_{ij}), and interlocking board/agency (A^{B}_{ij}) network edges.
- **Signature 4 — Asymmetric Hysteresis Loops ($K_{backward} < K_{forward}$):** When macroeconomic indicators or short-term fiscal subsidies temporarily rebound to pre-trigger levels, neither peripheral reproductive/career commitment (R_A, R_C) nor internal managerial fiduciary horizons (R_B) return along the forward trajectory. Re-pinning requires sustained safety cues ($T_{cover} > T_{recur}$ and intact Σ_{care}) across τ_{dev} exceeding the original trigger threshold.
- **Signature 5 — Intermittent Phase Slips Near the Bifurcation Boundary:** Unlike a smooth, monotonic response to an external shock, noisy coupled active rotators approaching or relaxing from a locking threshold exhibit characteristic **phase slips**—brief, punctuated desynchronisation bursts (Teramae & Tanaka, 2004; Goldobin & Pikovsky, 2005) alongside rising relaxation times (τ_{relax}).
- **Signature 6 — Bimodal / Anti-Phase Cluster Splitting:** Rather than shifting a unimodal population distribution smoothly in one direction, observables bifurcate into coexisting phase-separated clusters—such as simultaneous intensification of declarative N_0 orthodoxy alongside covert B_{sub} opportunity hoarding (the Chimeric Dual Strategy), or synchronised <Break> patch-foraging coexisting with defensive <Care> micro-enclaves.

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